

Project: Quantitative Comparison of Two Countries – Telling a Data-Driven Story

Objective

This final project challenges you to use real-world data to tell a meaningful, evidence-based story comparing two countries through a single indicator. You will apply the full range of tools and methods learned in this course—including data visualization, summary measures, probability distributions, hypothesis testing, etc.

The goal is to derive insights from the data—not merely to compute statistics, but to interpret and explain what these reveal about each country.

Part 1: Data Collection and Preparation

Data Selection – Framing the Story

- Choose a compelling socioeconomic or demographic indicator (e.g., income levels, life expectancy, educational attainment, etc) that allows you to uncover meaningful insights.
- Select two countries for comparison: one should be **Germany**.
- Briefly explain why the chosen indicator and these two countries provide an interesting narrative.
- Clearly cite the sources of your datasets.

Data Description – Setting the Scene

- Describe the chosen indicator and why it is significant in comparing these two countries.
- Specify:
 - **Data source**
 - **Sample size**
 - **Time period covered**
- Present relevant **summary statistics** for each dataset:
 - Mean, median, standard deviation, interquartile range, minimum, and maximum.
- Highlight any notable patterns or questions raised by these initial statistics.

Data Processing

- Clean and/or preprocess the data if necessary.
- Document any data transformation steps:

- What you did
- Why you did it
- How it affected the dataset

Data Visualization

- Select and create appropriate plots, histograms and kernel density plots for each country's dataset.
- Overlay the plots for a clear visual comparison of distribution shapes.
- Use proper labels, color coding, and legends to enhance interpretability.

Part 2: Empirical and Quantitative Distribution Comparison

Descriptive Statistics

- Calculate and interpret:
 - Mean, median, variance, standard deviation, skewness, and kurtosis.
- Compare the distributions in terms of symmetry, spread, and outliers.

Quantitative Comparison of PDFs

- Plot and interpret the probability density functions (PDFs) for each country.
- Use at least one of the following methods to quantitatively compare the distributions:
 - Kullback-Leibler Divergence
 - Earth Mover's (Wasserstein) Distance
 - Q-Q plot analysis
 - Any other appropriate method
- Interpret what the comparisons reveal about the similarity or difference between the distributions.

Part 3: Distribution Fitting and Normality Testing

Fitting Normal Distributions

- Fit a normal distribution to each dataset.
- Overlay the fitted curves on the histograms.

- Report the estimated mean and standard deviation of the fitted distributions.

Normality Tests

- Conduct normality tests such as the **Shapiro-Wilk** or **Kolmogorov-Smirnov** test.
- Report the test statistics and **p-values**, and interpret them in relation to the null hypothesis.
- Discuss how the sample size may affect the results.
- Include **Q-Q plots** and comment on any observed deviations from normality.

Alternative Distribution Fit (Optional)

- If the normal distribution is a poor fit, propose and fit a more appropriate distribution (e.g., log-normal, exponential).
- Compare the goodness of fit to the normal distribution using both visual and statistical evidence.

Part 4: Central Limit Theorem (CLT) Assessment

Bootstrap Resampling

- Generate 1,000 resamples (with replacement) from each dataset.
- Compute the sample mean for each resample.

Distribution of Sample Means

- Plot histograms of the resampled means for each country.
- Report the **mean** and **standard deviation** of the resampled distributions.
- Compare them to the original (population) statistics.

Normality of Sample Means

- Perform a normality test on the sample means.
- Interpret whether the sample mean distribution is approximately normal.
- Reflect on what this tells us about the applicability of the Central Limit Theorem in this context.

Part 5: Hypothesis Testing Framework

Formulating Hypotheses

- Formulate testable hypotheses aligned with your narrative. **Example:**

- Null Hypothesis (H_0): The mean (or distribution) of the selected indicator is the same across both countries.
- Alternative Hypothesis (H_1): The mean (or distribution) differs between the countries.

Conducting Hypothesis Tests

- Select and conduct an (or more) appropriate hypothesis test(s) based on the data; Explain why that's a proper test.
- Report the **test statistic**, **p-value**, and your conclusion.
- Tie the results back into your overarching story.

Part 6: Final Reporting – Telling the Story

Your final report should clearly communicate the findings and insights derived from your analysis. It should include:

- A summary of the key differences (or similarities) between the two countries
- Whether the normal distribution was appropriate
- What the resampling showed about distribution stability and central tendency
- If and how the Central Limit Theorem applied (Optional in story telling)
- The results of your hypothesis testing and their implications and interpretations
- Any surprising or notable discoveries

Submission Instructions

- **Team Formation:** Form a team of **4 to 5 students**.
- **Submission Format:** Only **one team member** should upload the following:
 - The **presentation/slides** (PPT, PPTx, PDF, or ODP)
 - The **final report (doc/docx/odt/tex and pdf)**
 - The **code files**, with proper documentation indicating:
 - Who wrote each part
 - What each file does
 - Where each code snippet is used in the report

- **Other team members** should submit a brief document including: Team member names, student ID numbers, the name of the person who submitted the final files

Code Documentation Tips

- Make sure each code file is well-commented with clear instructions for running it.
- Consider using platforms like GitHub or Google Colab for easy access and sharing.
- Include metadata (title, author names, file purpose) at the top of each script.

Helpful Tips for Creating Slides and Reports

- A clear, well-structured report/slide with logical sections and headings
- All relevant plots and test outputs, each with captions, descriptions, and interpretation (they should be referenced in the text)
- Clearly defined hypotheses and conclusions
- Proper citations for all data sources, code, and references used